Institute of Technology of Cambodia

Department Of Industrial and Mechanical Engineering

Lecture: **Dr. SRY VANNEI** (Course)
Dr. SIV Easeng (TP)

Subject: Strength of Materials
Group: I3-GIM-A1

Student:	**ID:**
1. CHAO SIVCHHUONG	e20220409
2. CHAT CHANPARADY	e20221655
3. CHEA SENGTRY	e20220606
4. CHEN CHANRAKSMEY	e20220632
5. CHENG CHENVICHET	e20220655
6. CHHAI LINDA	e20221092

Academic Year: 2024-2025

Type of Experiments

1. **Tensile Test**
2. **Three-Point Bending Flexural Test**
3. **Torsion Test**

1.Tensile Test

Table of Contents

I. Introduction .....
II. Objective .....
III. Theory .....
IV. Materials and Methods .....
V. Results and Discussion .....
VI. Conclusion .....

Tensile Test

I. Introduction

Mechanical testing plays an important role in evaluating fundamental properties of engineering materials, developing new materials, and controlling the quality of materials for use in design and construction. If a material is to be used as part of an engineering structure that will be subjected to a load, it is important to know that the material is strong enough and rigid enough to withstand the loads that it will experience in service. As a result, engineers have developed several experimental techniques for mechanical testing of engineering materials subjected to tension, compression, bending, or torsion loading. The most common type of test usedto measure the mechanical properties of a material is the tensile test. Tension test is widely used to provide basic design information on the strength of materials and is an acceptance test for thespecification of materials. The major parameter that describes the stress-strain curve obtained during the tension test are the tensile strength, yield strength or yield strain, Young's modulus, percent elongation, and the reduction in cross-sectional area. The tension test could also determine the stress concentration in this experiment.

II. Objective

1. To investigate the tensile properties (yield strength, ultimate tensile strength, elongation, fracture strength) of steel
2. To investigate the tensile properties (yield strength, ultimate tensile strength, elongation, fracture strength) of aluminum alloy
3. To investigate what is different in the tensile properties of steel and aluminum alloy

III. Theory

If a load is static or changed relatively slowly with time and is applied uniformly over a cross section /surface of a member, the mechanical behavior may be ascertained by a simple stress-strain test. These tests are most commonly conducted for metals at room temperature. There are three principal ways in which the load may be applied: tension, compression, and shear. Tension is one of

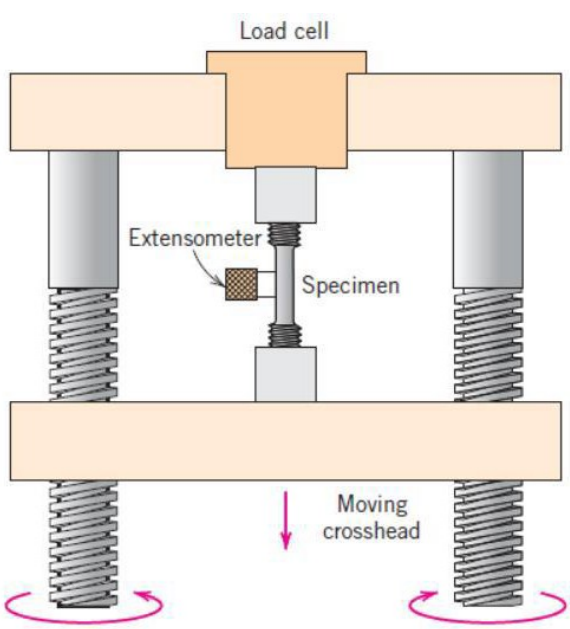
the most common mechanical stress-strain tests. A standard specimen is deformed, usually to fracture, with a gradually increasing tensile load that is applied uniaxially along the long axis of a specimen.

- An axial force applied to a specimen of original length (l_o) elongates it, resulting in a reduction in the cross-sectional area from A_o to A until fracture occurs.
- The load and change in length between two fixed points (gauge length) are recorded and usedto determine the stress-strain relationship.
- A similar procedure can be adopted with a sheet specimen.

By the way, Normal stress is obtained by dividing the load by the original area of the specimen's cross-section.

Nominal Stress $\sigma = \dfrac{Load}{initiaL\ Cross{-}sectional\ area} = \dfrac{P}{A_o}$

Nominal strain $\varepsilon = \dfrac{Elongation}{Initial\ gage\ Length} = \dfrac{\Delta l}{l_o}$



While the true stress σ_t and strain ε_t are given by:

$$\boldsymbol{\sigma_t = \frac{P}{A} = \frac{P}{A_o l_o} = \sigma(1+\varepsilon)}$$

$$\varepsilon_t = \int_{l_o}^{l} \frac{dl}{l} = ln \frac{l}{l_o} = ln\,(1+\varepsilon)$$

Where $l = l_o + \Delta l$

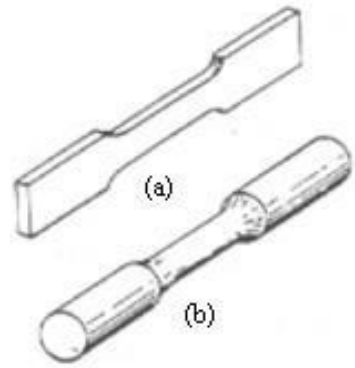
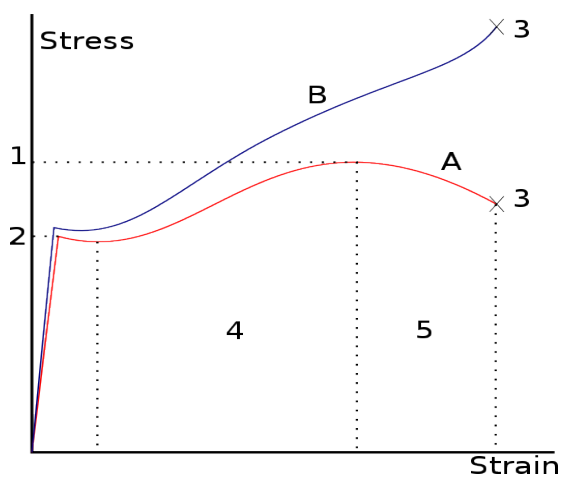


Fig.1 Common specimen shape of tensile test



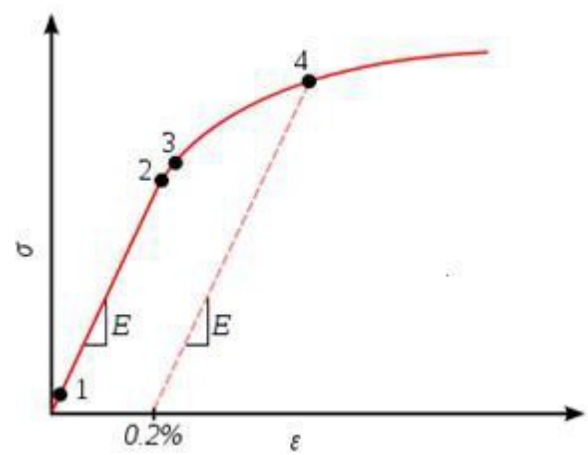


Fig. 1 Typical stress-strain diagrams of mildsteel

1. Ultimate Strength
2. Yield strength
3. Rupture
4. Strain hardening region
5. Necking region

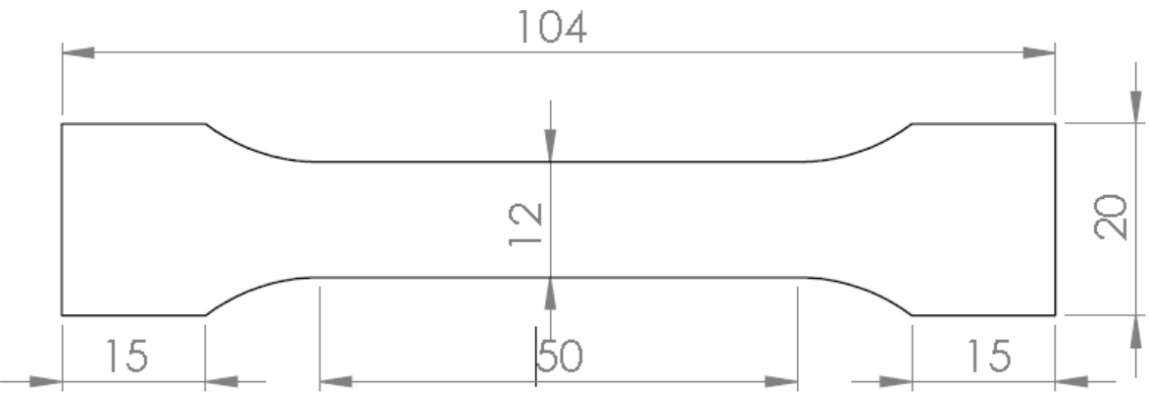
A: Nominal stress (F/A_0)B: Actual stress (F/A)

Fig.3 Typical engineering stress-straindiagram of non-ferrous alloys

1. True elastic limit
2. Proportional limit
3. elastic limit
4. Offset yield strength

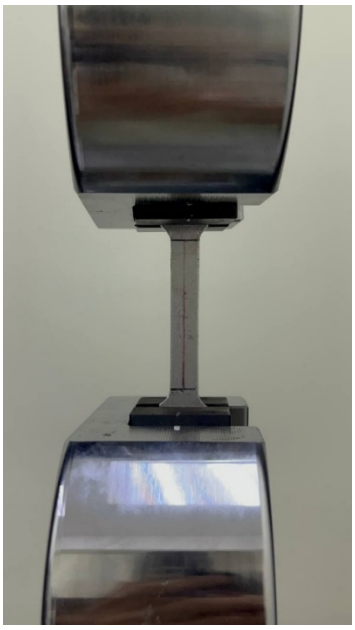
IV. Materials and Methods

The materials used in the experiment were steel and aluminum alloy, which were prepared in the form of a d using the ASTM E-8 standard. The tensile tests were performed by aSHIMADZU testing machine under a load of the average tensilevalues were selected to evaluate the tensile properties.

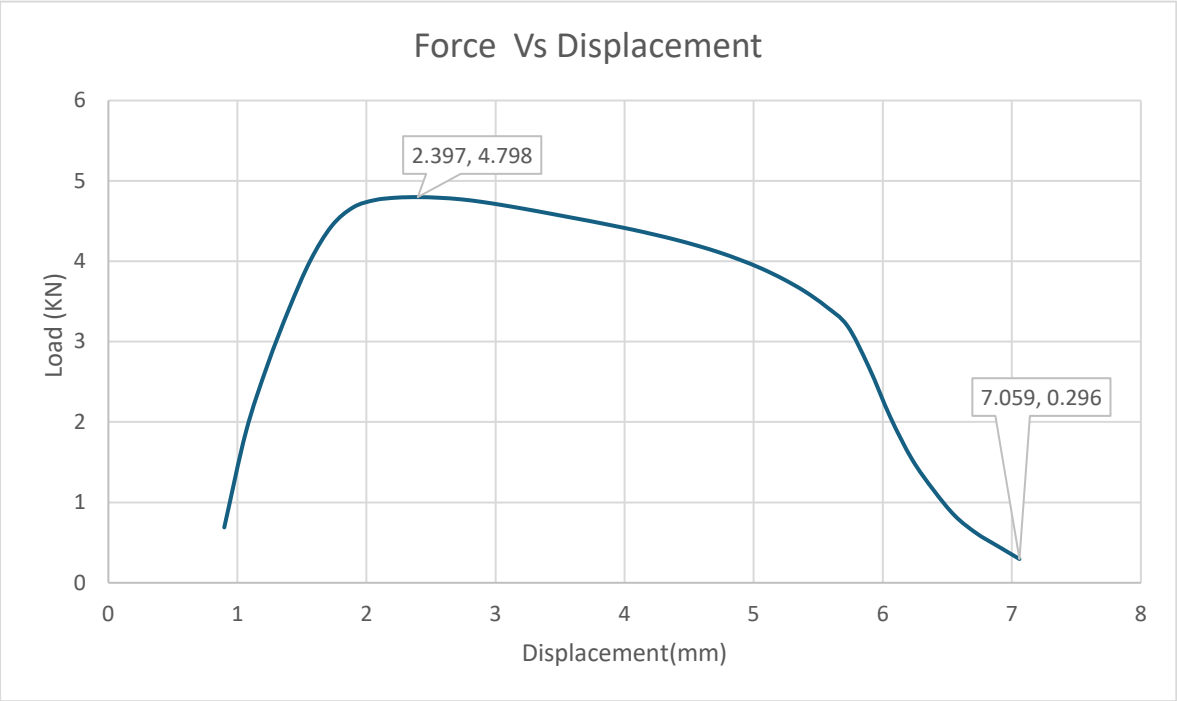


V. Result and Discussion

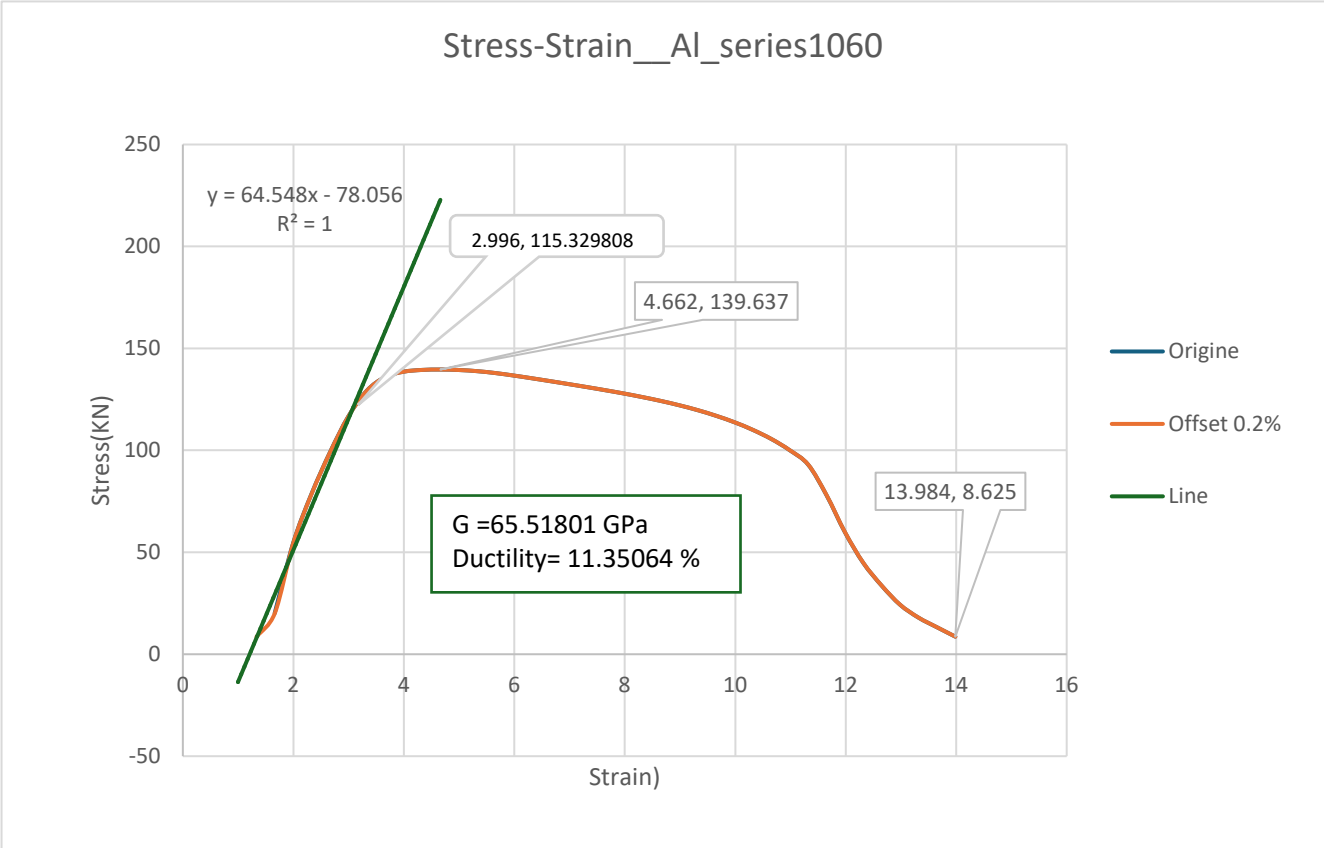
❖ Aluminum Series 1060



- **Diagram of Force Vs Displacement**



- **Diagram of Strain Vs Stress**

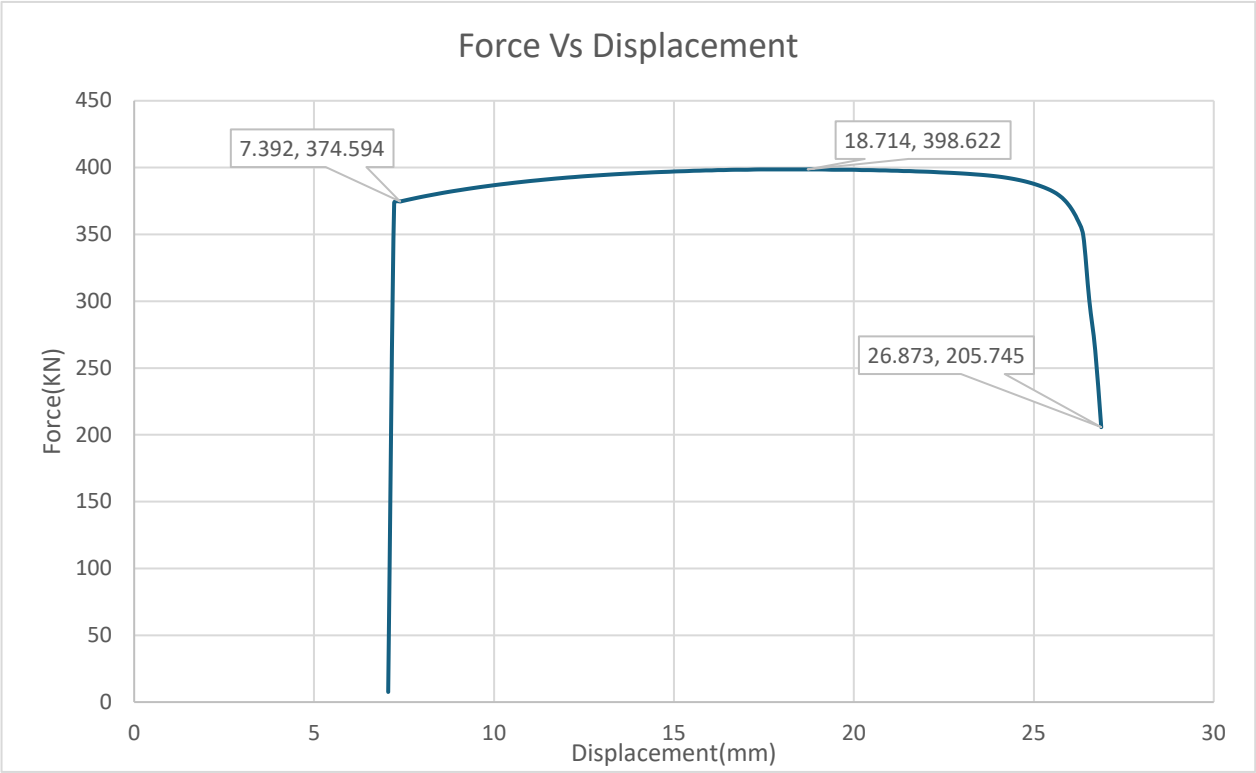


The result that we get:

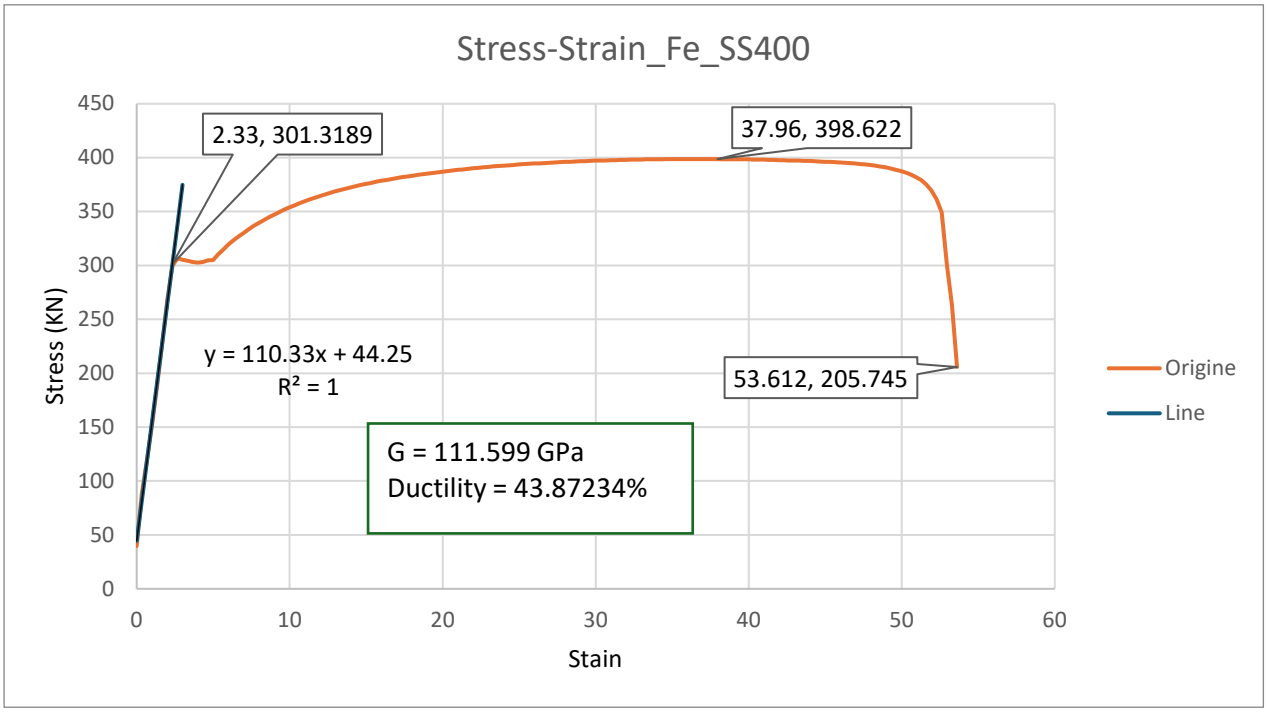
Al series 1060			
Type	**Experiment**	**Theory**	**Error**
G (GPa)	65.5180072	68	3.649989
Max Stress (Mpa)	139.637	130	7.413077
Yield Stress (Mpa)	115.329808	110	4.84528
Fractural Stress (Mpa)	7.115		
Ductility (%)	11.35064935		

❖ Carbon Steel SS400

- **Diagram of Force Vs Displacement**



- **Diagram of Stress Vs Strain**



- The result that we get:

Fe SS400			
Type	**Experiment**	**Theory**	**Error**
G (GPa)	111.5987973	190	41.26379
Max Stress (Mpa)	398.622	510	21.83882
Yield Stress (Mpa)	301.319	235	28.22085
Fractural Stress (Mpa)	205.745		
Ductility (%)	43.87234043		

Observation and comparison between the Fe SS400 and Al series 1060 after the test:

Fe SS400

Al series 1060

Difference Between Theoretical and Actual Discharge

We observe significant discrepancies between theoretical and actual discharge values from the experimental results. For example, Fe SS400 shows a large deviation of 41.27%, while Al series 1060 shows a smaller deviation of 3.65%. Moreover, the errors of the maximum stress, yield stress of the Al series 1060 are greater than Fe SS400. These differences arise due to several factors, such as measurement errors, specimen preparation, cutting techniques, depth of grip, and the testing machine used.

- **Specimen Geometry**: The specimens used were not perfectly symmetrical, which can affect how forces are distributed during testing.
- **Cutting Process**: During cutting, residual stresses and internal loads may be introduced into the specimens.
- **Measurement Errors**: Inaccuracies in measuring the original length (l_0), width (w), and thickness (d_0) using calipers or manual methods can introduce errors.
- **Machine Setup**: The test speed was set at 1 mm/min. However, improper alignment of specimens in the machine can cause uneven force application, combining shear and tensile forces.
- **Gripping Depth**: Unequal clamping depth at the upper and lower ends of the specimen may lead to inconsistent loading and premature failure.

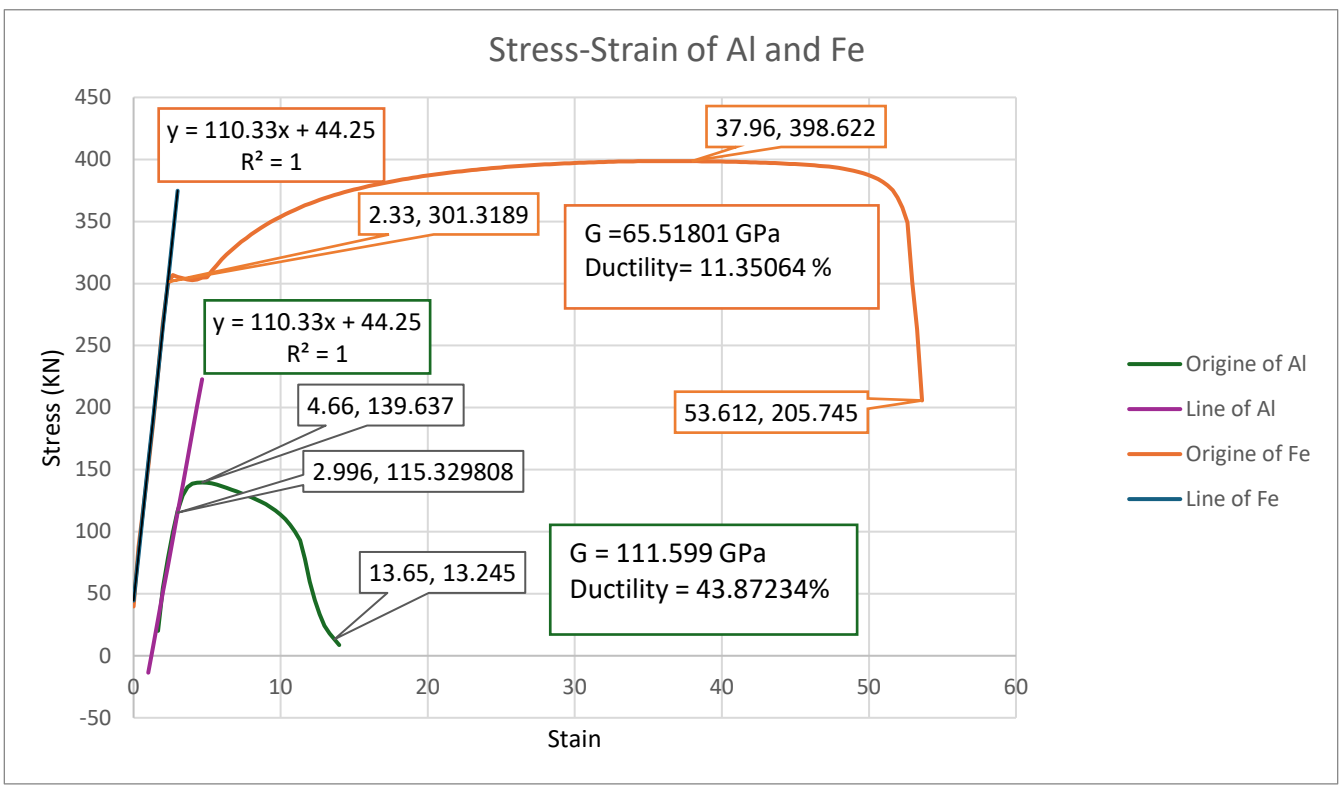
- **Stress stored in the material**: That stress occurred when we cut, mill the material.

Suggestions for Improvement

To increase the accuracy and reliability of the experiment, we recommend the following improvements:

- **Stress Relief**: Heat-treat or stress-relieve specimens before testing to eliminate residual stresses.
- **Precision Measurement**: Use high-precision instruments with minimal tolerance, and take multiple readings to ensure accurate dimensions.
- **Symmetrical Clamping**: Ensure that both the upper and lower clamping depths are equal to maintain uniform loading.
- **Use of CNC**: If possible, fabricate specimens using CNC machines to ensure high precision and symmetry.
- **Machine Calibration**: Check and calibrate the tensile testing machine regularly. Also, choose an appropriate testing speed based on the material's properties.
- **Release Stress**: By heating the specimens to a temperature that we set.

The difference between the tensile properties of steel and aluminum alloy



Property	Steel SS400	Aluminum Alloy 1060 series
Max Strength (MPa)	398.622	139.637
Yield Strength (MPa)	301.319	115.329808
Ductility (%)	43.87234043	10.1461039
Elastic Modulus (GPa)	111.5987973	65.5180072
Density (g/cm^2)	7.85	2.7
Fracture Stress (MPa)	205.745	13.245

By this graph and this Data, we can say that:

The mechanical properties of Fe SS400 are better than Al series 1060, including Maximum strength, Yield Strength, Ductility, and Elastic Modulus. However, **Aluminum Alloy 1060** has a much **lower density**, making it preferable in weight-sensitive applications like aerospace or automotive components.

VI Conclusion

From the tensile test experiment, it is clear that both material properties and testing conditions significantly influence the accuracy and outcome of the results.

Experimental Accuracy and Error:

The observed differences between theoretical and actual discharge values, such as the 41.27% deviation for Steel SS400 and the 3.65% deviation for Aluminum 1060, highlight the impact of various experimental errors. These include asymmetrical specimen geometry, internal stress from cutting, measurement inaccuracies, misalignment during machine setup, and unequal gripping depth. Addressing these factors is essential to improve the consistency and reliability of tensile test results.

Material Property Comparison:

The tensile properties of Steel SS400 and Aluminum 1060 show clear differences:

- **Steel SS400** has higher ultimate strength (398.62 MPa), yield strength (301.319 MPa), ductility (~43.87%), and fracture resistance (678.4 MPa), making it suitable for high-strength applications.
- **Aluminum 1060**, though lower in strength, offers advantages in ductility (~10.15%) and significantly lower density (2.7 g/cm³), making it ideal for lightweight applications where flexibility and corrosion resistance are important.

Improving the test

To increase the accuracy and reliability of the experiment, we recommend the following improvements:

- **Stress Relief**: Heat-treat or stress-relieve specimens before testing to eliminate residual stresses.
- **Precision Measurement**: Use high-precision instruments with minimal tolerance, and take multiple readings to ensure accurate dimensions.
- **Symmetrical Clamping**: Ensure that both the upper and lower clamping depths are equal to maintain uniform loading.
- **Use of CNC**: If possible, fabricate specimens using CNC machines to ensure high precision and symmetry.
- **Machine Calibration**: Check and calibrate the tensile testing machine regularly. Also, choose an appropriate testing speed based on the material's properties.
- **Release Stress**: By heating the specimens to a temperature that we set.

The mechanical properties clearly show that **Steel SS400** is superior in terms of strength, stiffness, ductility, and toughness compared to **Aluminum 1060 series**. However, aluminum's low density gives it an advantage in applications where **weight reduction** is more important than mechanical strength.

VII Reference

1. ASTM A36 (SS400, S275) Structural Carbon Steel :: MakeItFrom.com
2. 1060 (Al99.6, A91060) Aluminum :: MakeItFrom.com
3. TP-RDM 2023-2024.pdf
4. Torsion Test.pdf
5. Comprehensive Guide to JIS G3101 SS400 Steel: Properties, Equivalents, and Applications - MFG Shop

2.Three-Point Bending Flexural Test

Table of Contents

I. Introduction.....
II. Objective .....
III. Theory .....
IV. Materials and Methods .....
V. Results and Discussion .....
VI. Conclusion .....
VII. Reference .....

Three-Point Bending Flexural Test

I. Introduction

Three-point bending tests are used to evaluate the mechanical properties of materials, specifically their flexural strength and modulus of elasticity. This involves applying force to the midpoint of a sample that is supported at two ends, causing it to bend. The results provide essential data for understanding how materials will perform under bending stresses, which is crucial for designing safe and reliable structures.

II. Objective

1. Determine Flexural Strength: Measure the maximum stress a material can withstand in bending
2. Assess Modulus of Elasticity: Calculate the material's stiffness.
3. Evaluate Ductility and Brittleness: Understand the material's deformation before breaking.
4. Analyze Failure Modes: Identify how and where the material fails.
5. Guide Material Selection: Provide data for selecting suitable materials for structural applications

III. Theory

Flexure Formula: $\sigma = -(M_{max} * y/I)$

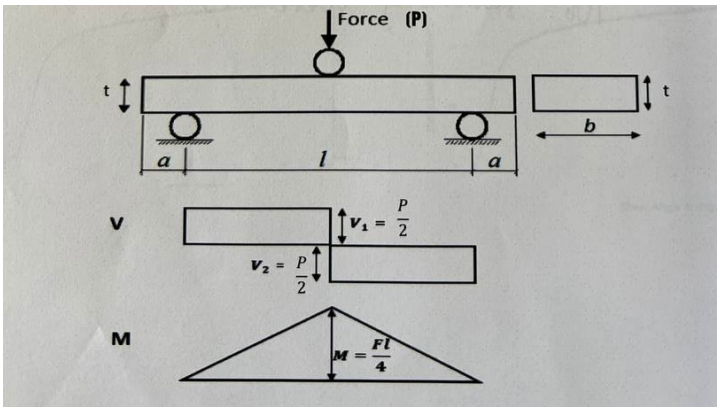
Bending Moment Varies from zero at the support to maximum ($P*L/4$) at the center:

By $y = t/2$

$I = (3PL)/(2bt^2)$

So that, $\sigma = (3PL)/(2bt^2)$

$€ = (6t\sigma)/L$



The test method for conducting the test usually involves a specified test fixture on a universal testing machine. Details of the test preparation, conditioning, and conduct affect the test results. The sample is placed on two supporting pins a set distance apart.

Calculation of the flexural stress σ_f

$$\sigma_f = \frac{3FL}{2bd^2}$$ for a rectangular cross section

$$\sigma_f = \frac{FL}{\pi R^3}$$ for a circular cross section[1]

Calculation of the flexural strain ϵ_f

$$\epsilon_f = \frac{6Dd}{L^2}$$

Calculation of flexural modulus E_f[2] /mm)

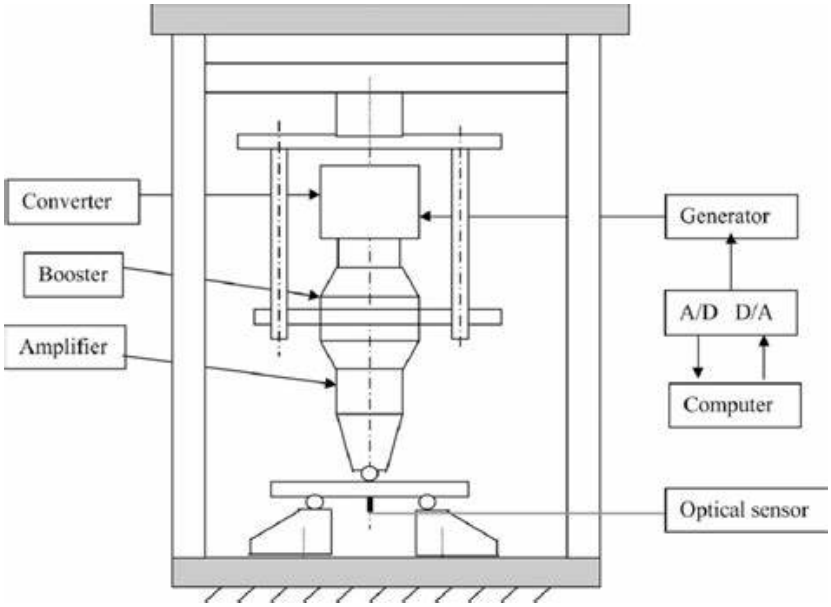
$$E_f = \frac{L^3 m}{4bd^3}$$

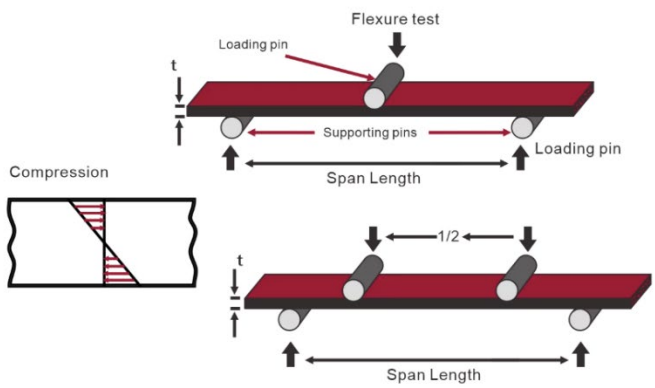
IV. Materials and Methods

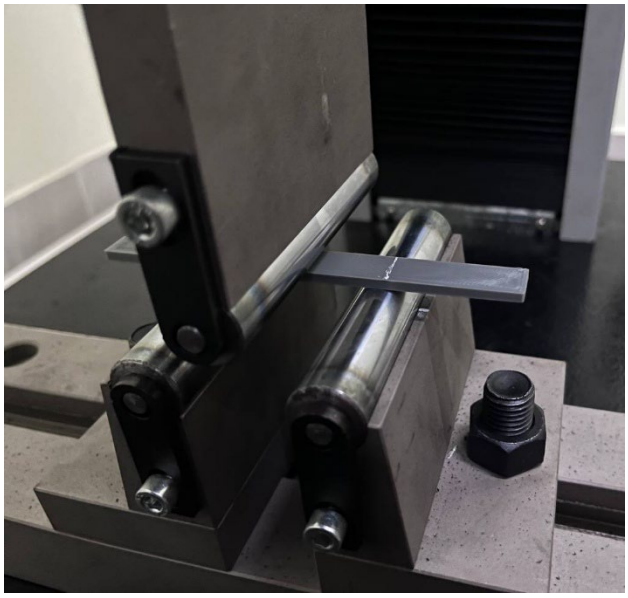
The test method for bending Strength is usually ASTM D790 or ISO 178. ASTM generally uses 3.4mm(thick) x 13.2mm(wide) x 127.1mm(long) (0.1271″ x 0.0034″ x 0.0132″) plastic strip samples. In contrast, ISO uses 4mm x 10mm x 80mm strip samples. During the test, the sample is placed on two fixed anvils at a certain distance and then pressed at the center of the two fulcrums with a point (three-point test method) from top to bottom at a speed of 1 mm/min until the sample breaks or exceeds a certain limit. But we take L0 to be 50 mm only.

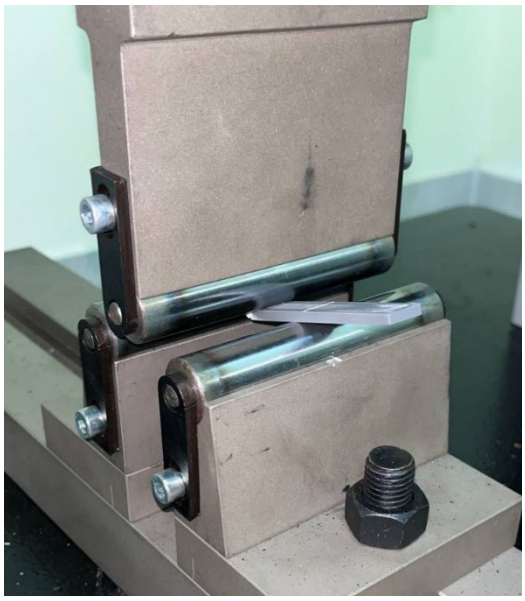
The data of the specimen that we use in this test:

w	13.2
t	3.4
L0	50
m	41.394

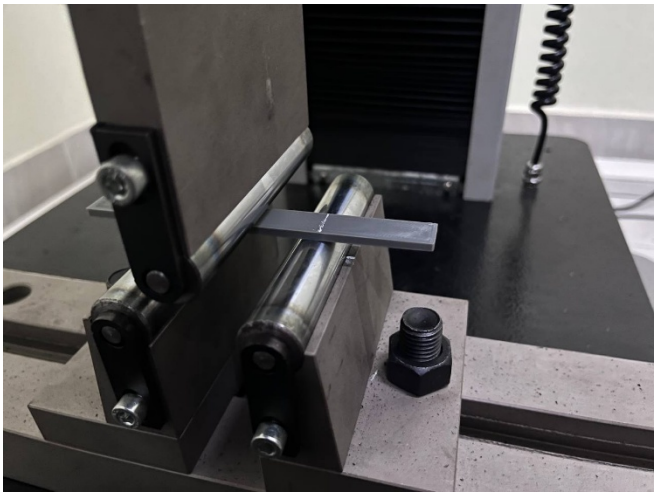

Converter
Booster
Amplifier
Generator
A/D D/A
Computer
Optical sensor


Flexure test
Loading pin
t
Supporting pins
Loading pin
Span Length
Compression
1/2
t
Span Length

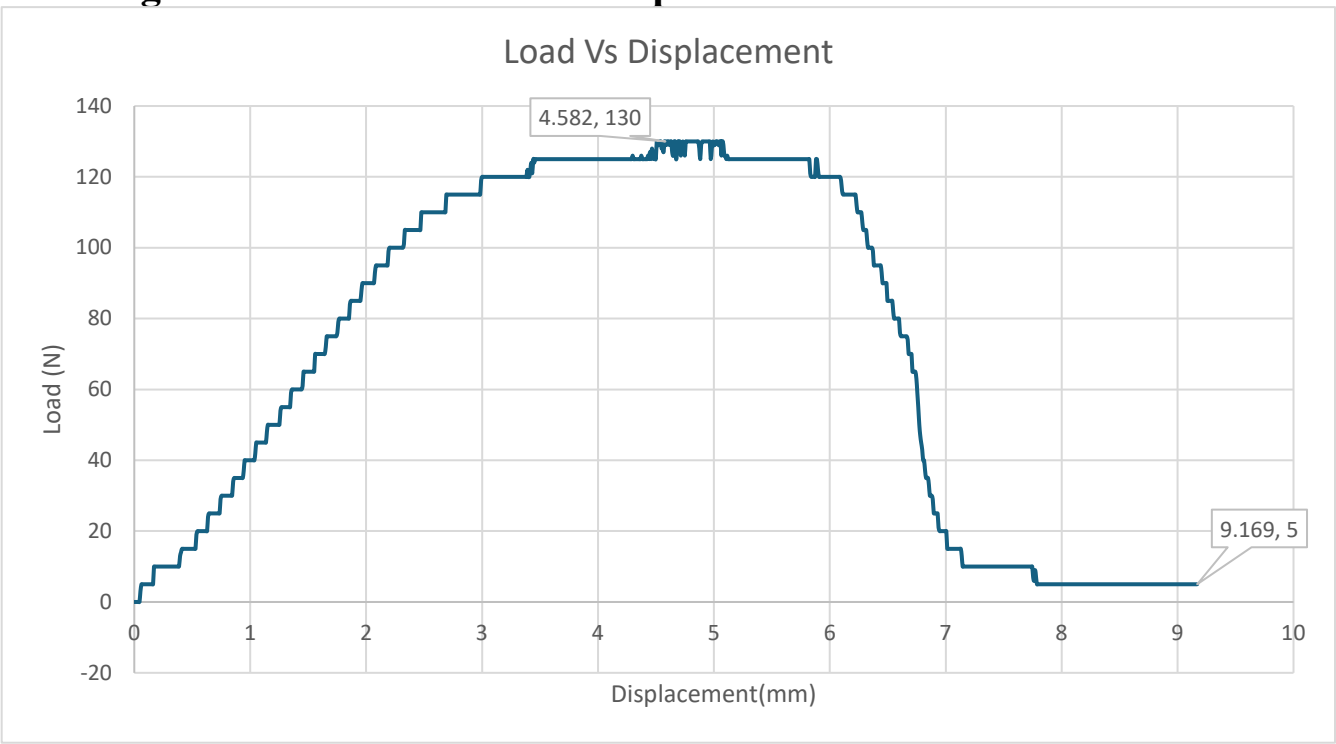




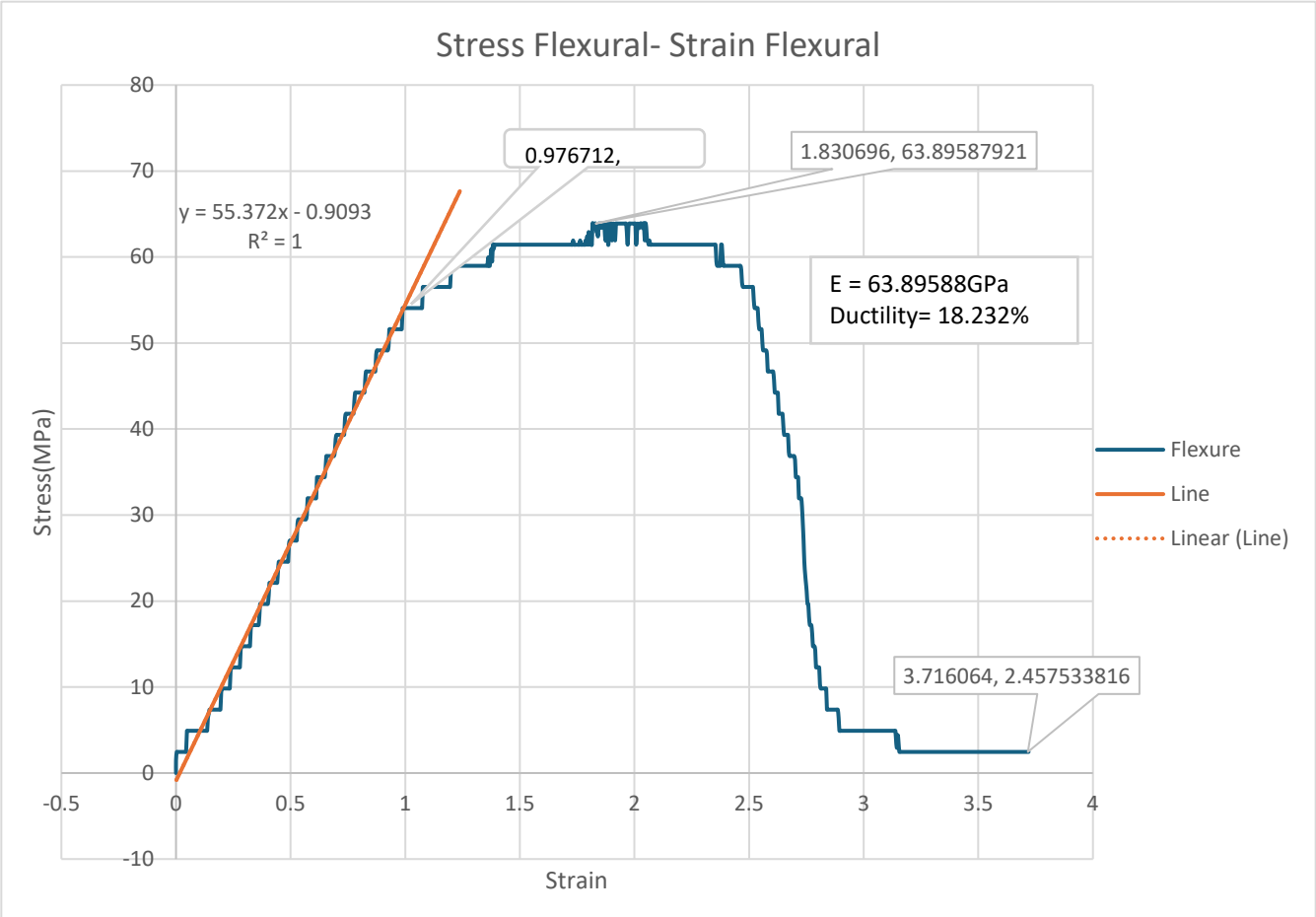
V. Results and Discussion



- **The diagram of the Load Vs the Displacement**



- **The diagram of the Flexural Stress Vs Flexural Strain**



The result that we get from the test:

Plastic ABS			
Type	**Experiment**	**Theory**	**Error**
Flexural Strength (MPa)	63.8958792	75	14.80549439
Flexural Modulus (MPa)	2493.31262	2500	0.267495328
Yield Strength (MPa)	54.1220515	71.7054758	24.52173163
Yield Strain	0.976712		
Breaking force(N)	97		
Breaking stress (MPa)	3		
Proof force (N)	96.85413		
Proof stress (MPa)	2		
Ductility (%)	18.232		

Evaluate the result:

Based on the results, we observed some deviations in the Flexural Strength (14.81%), Flexural Modulus (0.27%), and Yield Strength (24.52%). These discrepancies may be attributed to several possible factors, including the specimen, measurement process, equipment, and setup:

- **Specimen**: The 3D printing machine used to produce the specimen may have imperfections or calibration errors, affecting the quality of the specimen.
- **Measurement**: Inaccuracies could arise from the measuring tools (e.g., calipers) or due to a lack of precision in reading and recording dimensions.
- **Equipment**: The 3-point bending test machine itself may influence the results if not properly calibrated or maintained.
- **Tool Setup**: Misalignment of the specimen, such as not being centered or symmetrically placed during the test, could significantly impact the accuracy of the measurements.
- **Temperature**: Temperature could change the density of the material.

Improving the test:

- Check the 3D print machine to make sure that the machine can print with as much tolerance as possible
- Measurement: know the tolerance that was made by the machine and try to measure many times and compare with your teammate.
- Equipment: Check the machine to make sure the specimens that we place on the machine are constant without vibration and stay at the middle point of the specimens

The Brittleness and ductility on the test:

- The Brittleness is none, since the ABS plastic doesn't have.
- The Ductility is 18.232%, so it is a good and higher ductility, and the yield strain suggests the material can deform significantly before failure, indicating good ductility and lower brittleness.

Ductility: 18.23% → High

Yield Strain: 0.976712%

Brittleness Index (BI) = 54.122 / 0.976712 ≈ 55.4 (moderate)

ABS is ductile, not brittle. It can undergo significant deformation before failure, meaning it won't shatter or crack suddenly. The high ductility confirms that the material can absorb energy and deform plastically — ideal for impact-prone applications.

Analyze Failure Modes: Identify how and where the material fails:

Failure mode refers to how and where the material fails — brittle fracture, yielding, necking, or breaking under tension/compression.

Breaking Force: 97 N

Breaking Stress: 3 MPa

Proof Force: 96.85 N

Proof Stress: 2 MPa

Yield Strength: 54.12 MPa

The material shows clear yield behavior before breaking, indicating a ductile failure mode. It resists elastic deformation up to 54 MPa, then undergoes plastic deformation before eventually breaking at much lower stress (3 MPa). This gradual failure is safe and predictable, a key trait of engineering-grade plastics like ABS.

Guide Material Selection:

Use mechanical properties to determine if ABS is suitable for your application (e.g., housing, load-bearing parts, 3D printing).

- **Good Flexural Strength**: Withstands bending
- **Good Elastic Modulus**: Resists deformation
- **High Ductility**: Can deform safely
- **Ductile Failure**: No sudden fracture
- **Moderate Errors**: Acceptable for practical use

ABS is a strong candidate for structural, mechanical, and functional parts where toughness, impact resistance, and flexibility are needed.

Evaluation Area	Result
Flexural Strength	Suitable (63.9 MPa)
Stiffness	Good (2759 MPa)
Ductility	High (18.23%)
Brittleness	Low (Not brittle)
Failure Mode	Ductile
Material Suitability	Very suitable for structural applications

VI. Conclusion

Based on the experimental results and analysis, ABS (Acrylonitrile Butadiene Styrene) demonstrate mechanical properties that make it highly suitable for structural applications. **The flexural strength (63.9 MPa), flexural modulus (2493.31262 MPa)**, and yield strength (54.12 MPa) align well with the theoretical values and industry standards for ABS. The material exhibits **high ductility (18.23%)** and a **yield strain of 0.9767%**, confirming that it can undergo significant plastic deformation before failure. This indicates a **ductile** failure mode rather than a brittle one, which enhances its reliability in real-world applications.

Improving the test:

- Check the 3D print machine to make sure that the machine can print with as much tolerance as possible
- Measurement: know the tolerance that was made by the machine and try to measure many times and compare with your teammate.
- Equipment: Check the machine to make sure the specimens that we place on the machine are constant without vibration and stay at the middle point of the specimens

Although minor deviations in flexural strength and yield strength were observed, these are within acceptable error margins and likely due to equipment or measurement limitations. Overall, ABS proves to be a **durable, tough, and flexible material** ideal for use in mechanical components, 3D printing, and load-bearing parts where both strength and deformation resistance are critical.

VII. Reference

1. Flexural strength and flexural modulus of injection molding plastics
2. Acrylonitrile Butadiene Styrene (ABS) | ECK Plastic Arts | Plastic Fabrication, Thermoforming, Assembly - Binghamton, NY
3. Three-point flexural test - Wikipedia

3.Torsion Test

Table of Contents

I. Introduction.....
II. Objective .....
III. Theory .....
IV. Materials and Methods .....
V. Results and Discussion .....
VI. Conclusion .....
VII. Reference .....

Torsion Test

I. Introduction

Torque is a moment that tends to twist a member about its longitudinal axis. During the torsion test, a sample is subjected to stress via a torsion moment (torque). Shear stress occurs in the sample with distort the sample. The distortion is expressed as the wasting of the two sample ends concerning each other. The most common application is provided by transmission shafts, which are used to transmit power from one point to another.

II. Objective

1. To study the torsional moment and angle of twist of steel
2. To investigate the shear stress and shear angle of steel
3. To investigate what is different about the torsional moment and angle of twist of steel
4. To investigate what is different about the shear stress and shear angle of steel

III. Theory

When an external torque is applied to a shaft, it creates a corresponding internal torque within the shaft. The relation of torque and shear stress is represented as follows:

$$\boldsymbol{\tau_{max}} = \frac{\boldsymbol{TC}}{\boldsymbol{J}}$$

Here

$\boldsymbol{\tau_{max}}$: the maximum shear stress in the shaft

T: the resultant internal torque acting at the cross-section***J:*** the polar moment of inertia of the cross-sectional area***C:*** the outer radius of the shaft

The angle of twist $\emptyset$ and the maximum shearing strain $\boldsymbol{\gamma_{max}}$ are related as follows:

$$\gamma_{max} = \frac{C\emptyset}{L}$$

$$\emptyset = \frac{TL}{JG}$$

Where G is the modulus of rigidity or shear modulus of the material

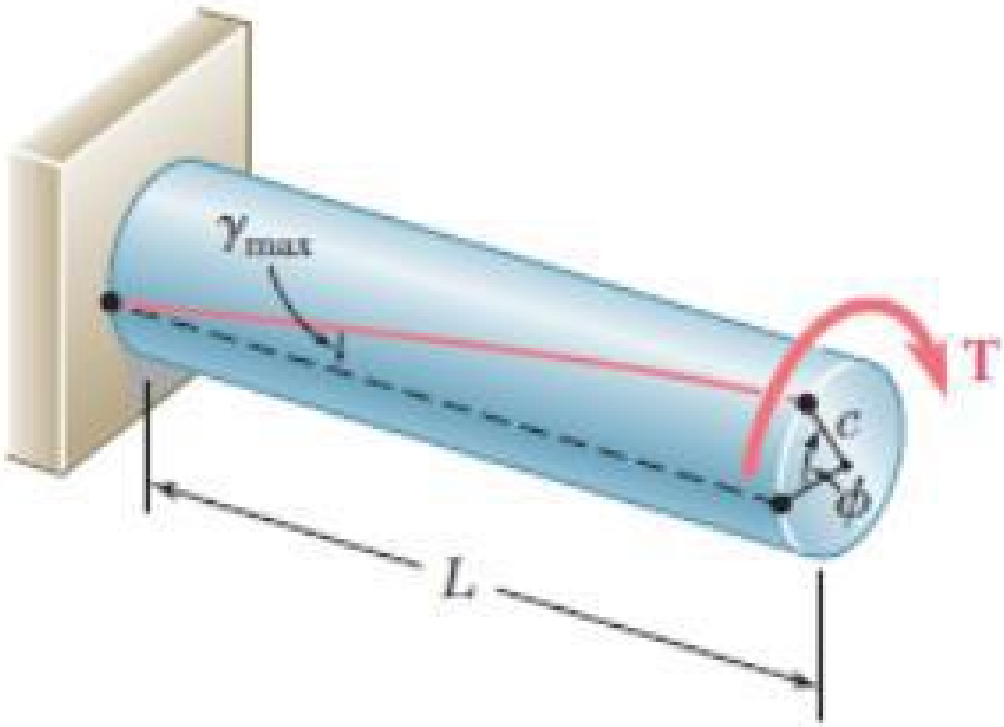


Figure 2. Shear stress and twist angle

We can determine the size of the shaft's cross-section using the torsion formula, provided the material behavior is linear elastic, as follows:

$$\frac{J}{C} = \frac{T}{\tau_{allow}}$$

- For a solid shaft, $J = \frac{\pi}{2} C^4$

I.V. Material and Methodology

• Material

The materials used in the experiment were steel and aluminum alloy, which were prepared as a bone shape in the rolling direction. The WP 510 testing machine performed the torsion tests under a torque of 200 kN. Two specimens are tested with the WP 510 testing machine.

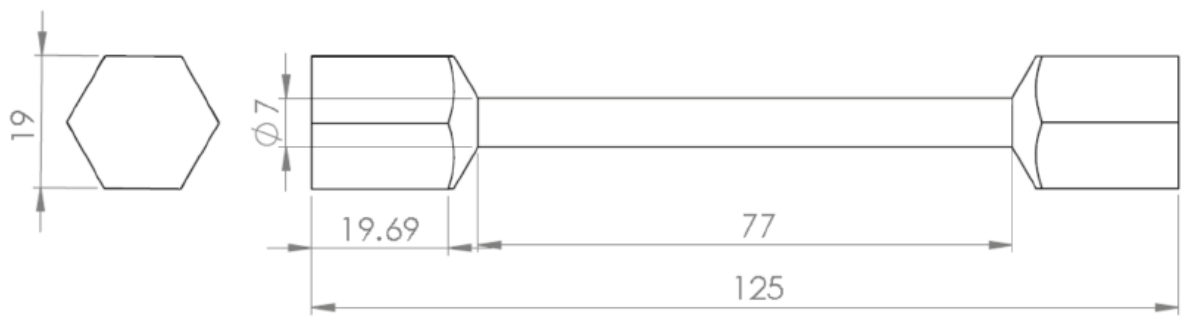


Figure 3. Torsion Specimen

• Methodology

For this test, we will use the machine WP 510 Torsion Tester.

The Instruction Manual of Torsion Test by using

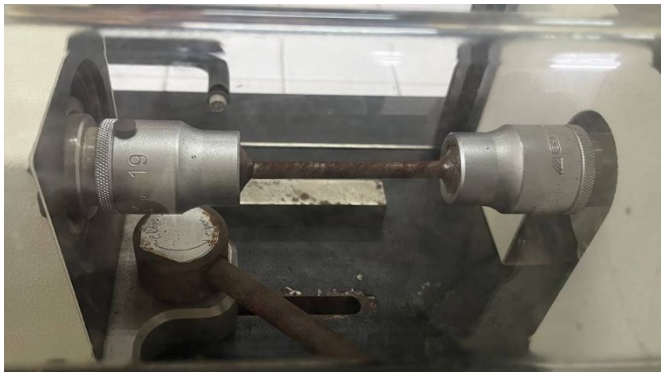
WP 510 Torsion Tester

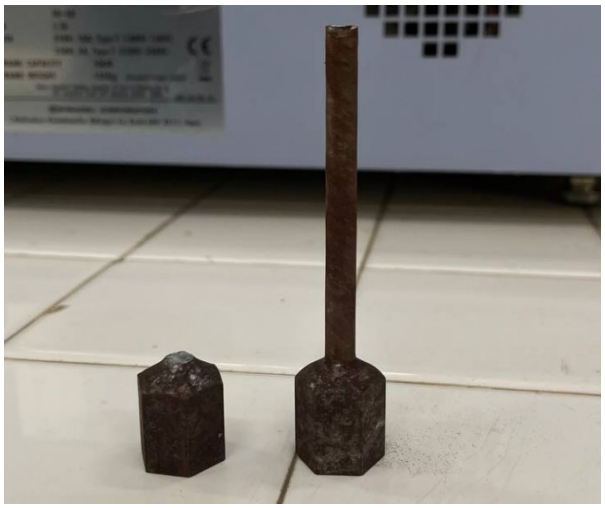
- Turn on the machine by rotating the button to ON
- Turn on the computer
  - Enter the program grunt WP510
- Specimen Assembly
  - Connect the specimen with the specimen holder (commercial 19mm socket)
  - Click on Tack Continuous Record or the button Start if the specimen cannot connect with the specimen holder
  - Press the button, Tara, before starting the experiment. Set the angle to Zero
  - Click on Tack Continuous Record or Press the button Start
  - Click on save when the specimen is fractured

- When does the experiment
  - Turn off the computer first
  - Turn off the machine by rotating the button OFF

IV. Results and Discussion

It is a result that we get from the machine WP 510 Torsion Tester.

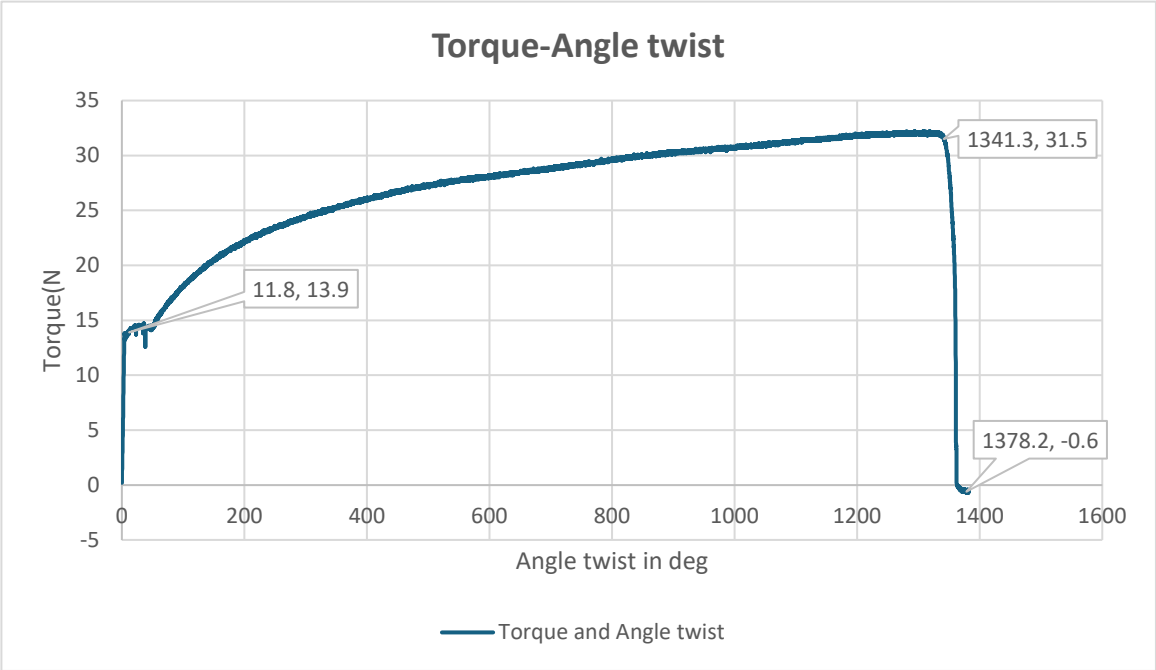




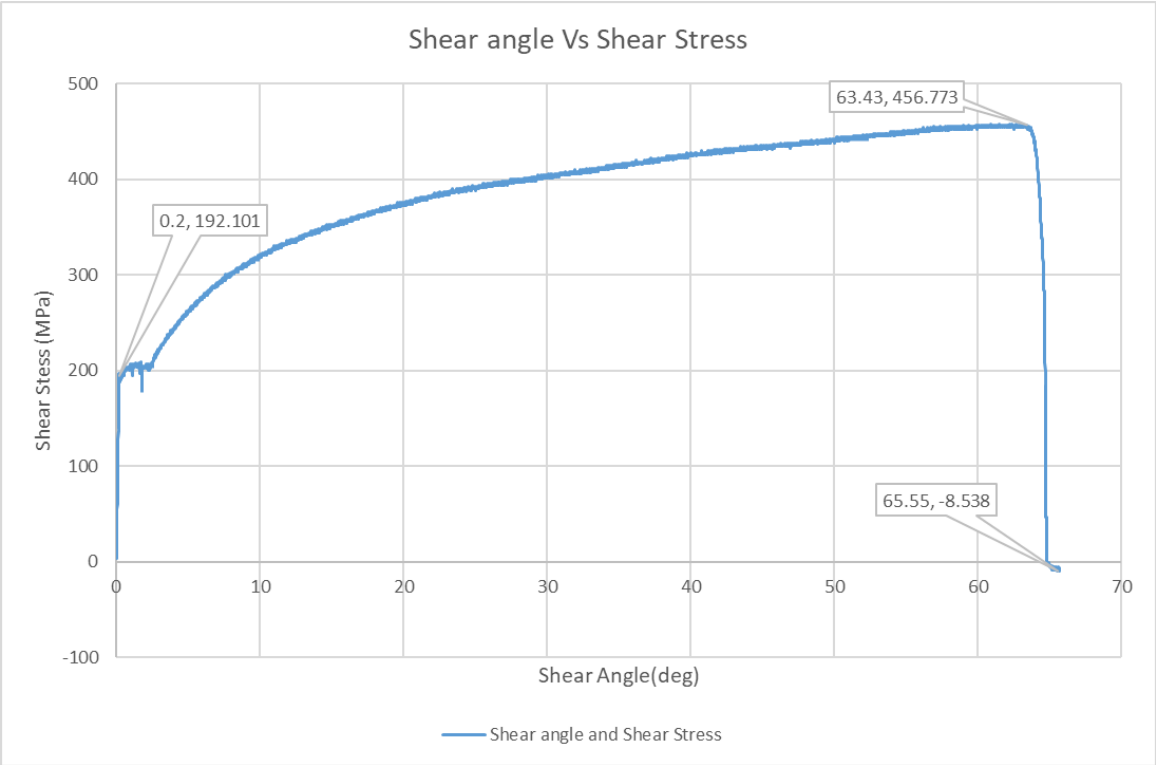
It is the dimension of specimens that we take to experiment.

We have:			
L(mm)	74.65	**L(m)**	0.07465
d(mm)	7.1	**d (m)**	0.0071
Ip(m^4)	2.49352E-10		
k	0.02		

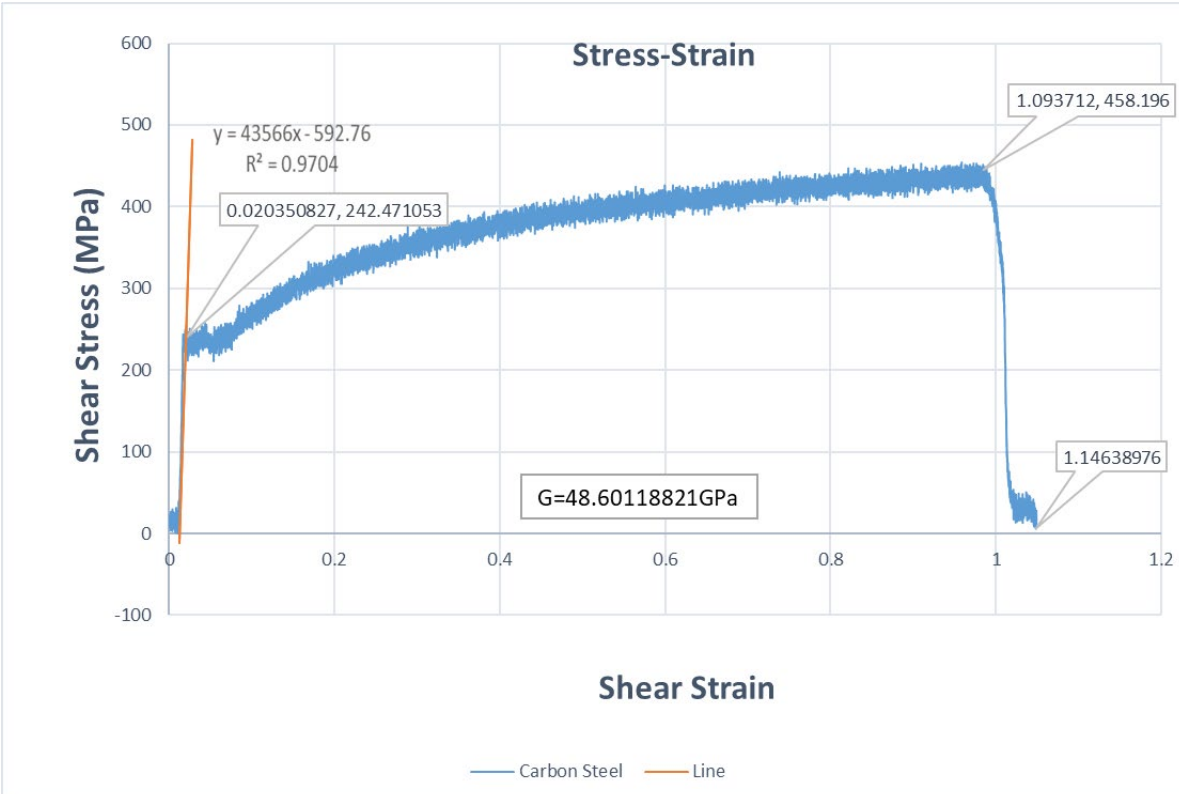
- **Diagram of Torsional Moment Vs Angle of Twist**



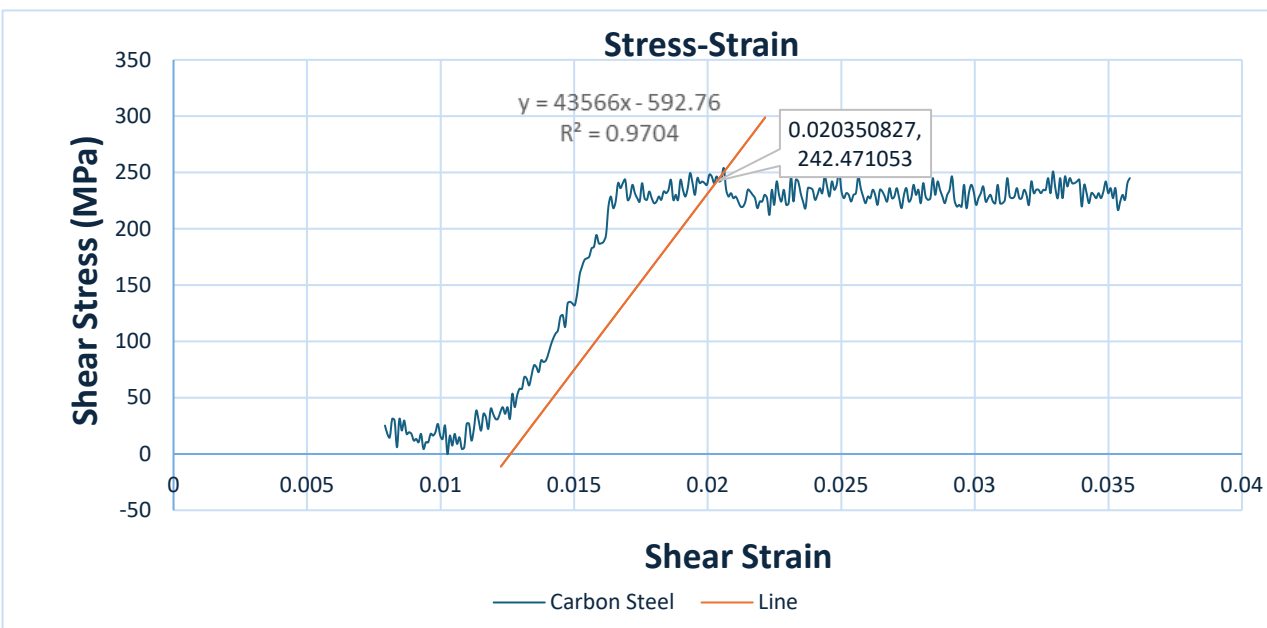
- **Diagram of Shear Angle Vs Shear Stress**



- **Diagram of Shear Strain Vs Shear Stress**



When we zoom in to see the intersection point between the linear line and the Stress-Strain curve:



The result that we get from the test:

Type	Experiment	Theory	Error
G (GPa)	48.60118821	79	38.47950859
Max Shear Stress (Mpa)	458.196	350	30.91314286
Yield Shear Stress (Mpa)	242.471053	250	3.0115788
Fractural Shaer Stress (Mpa)	7.115		

By this result, we have seen the value's error is 38.47%. It is so a big value, so it means it has many errors that happened when we experimented. After we discussed, we found some factors that make forward to big like this. The factors that we found are:

- **Specimen:** The dimension that is not fit the machine. It means the specimens that we take to test has big tolerance.
- **Machine:** The value of the angle of twist or torque doesn't fit our specimens because generally, we take the angle of twist 45 to 90 degrees, and it takes a long time.
- **The specimen** is not symmetric, and both sides
- **The stress** that is stored in the specimens since we have not heated or released stress before the test
- **Speed selection** that we use to apply to the test
- **The measure** is not exactly
- **The material** that we take or buy from the market to use in this test is not a carbon steel 100%. it means it is a carbon steel alloy, that why it was make a big error.

To improve the result and reduce the error in the next test, we should follow some steps:

- Buy and choose the real carbon steel
- Make the symmetrical specimens
- Heat the specimens to release the stresses before doing the test
- Choose the right speed that fits the material
- Measure and read many times to make sure the scale is correct

1. To study the torsional moment and angle of twist of steel

From the **Torque vs. Angle of Twist** graph, we can observe:

- As the angle of twist increases, the torque also increases linearly at first (elastic region).
- After a certain point, torque starts to plateau, indicating **plastic deformation**.
- The maximum torque is approximately **32 N·m**, after which the specimen **fails** or softens.
- This shows the **torsional behavior** of carbon steel: it can store elastic energy before yielding.

2. To investigate the shear stress and shear angle of steel

The **Shear Stress vs. Shear Angle** graph shows:

- A nonlinear relationship at higher angles, but nearly linear at the beginning.

- Shear stress increases with shear angle until it reaches a peak (~450 MPa), after which the material yields.
- This helps identify the **maximum shear strength** and how carbon steel behaves under shear loading.

3. To investigate what is different about the torsional moment and angle of twist of steel

The **Torsional Moment vs. Angle of Twist** curve:

- Is **initially linear** (Hookean behavior), showing a constant slope (proportional to shear modulus GGG).
- Shows **nonlinear plastic behavior** after the yield point.
- It helps in understanding how **torque leads to angular deformation**, and when the material starts to **permanently deform**.

Difference: Torque involves **circular (rotational) loading**, and the response is measured in terms of angle of twist and torsional stiffness.

4. To investigate what is different about the shear stress and shear angle of steel

The **Shear Stress vs. Shear Angle** graph:

- Also shows **initial linearity,** indicating elastic deformation.
- At higher angles, the curve deviates, indicating **plastic flow or yielding**.
- The final drop shows material failure.

V. Conclusion

From the experimental results and analysis, we observed a significant error margin of **38.47%**, which is considerably high. This suggests that multiple factors may have contributed to inaccuracies during the torsion test. After careful discussion, we identified several potential causes:

- **Specimen Issues**: The test specimens had dimensional tolerances that did not fit the testing machine properly, possibly affecting the results.
- **Machine Limitations**: The angle of twist applied (45–90 degrees) may not have been ideal for our specimen properties, leading to longer test times and potential misreadings.
- **Asymmetry**: Specimens were not perfectly symmetrical, which could introduce stress concentration and uneven deformation.
- **Residual Stress**: The specimens may have retained internal stresses due to a lack of proper heat treatment before testing.
- **Speed Selection**: Inappropriate testing speed may have caused inaccurate stress-strain responses.
- **Measurement Inaccuracy**: Instrument reading and human error during measurement also likely contributed.

- **Material Composition**: The steel used may not have been 100% pure carbon steel; if alloyed, this would impact its mechanical properties.

Recommendations for Improvement

To reduce the error and improve accuracy in future experiments, we suggest:

- Using **verified carbon steel** with known composition.
- Ensuring **symmetrical specimens** are machined precisely.
- **Heat-treating specimens** to release internal stresses before testing.
- Selecting the **appropriate testing speed** for the material type.
- **Careful and repeated measurements** for better reliability.
- Confirming the **material's properties and certification** before use.

Based on the torsion test results, the material exhibits **good shear strength and yielding behavior**, with yield shear stress closely matching the theoretical value. However, the **low shear modulus** and **very low fracture shear stress** suggest either **material softening, experimental error, or post-yield failure mechanisms**. The high maximum shear stress indicates that the material can withstand significant torsional loads before failure.

VI. Reference

1. Torsion testing experiment (instructor) | PDF
2. Modulus of Rigidity
3. Comprehensive Guide to JIS G3101 SS400 Steel: Properties, Equivalents, and Applications - MFG Shop
4. Table of material properties for structural steel S235, S275, S355, S420